

Homework 3

Quadratics and the discriminant · about 60 minutes

Each question graded N0–E8

Answers · tutor's working, not an official mark scheme

Show all working. Give answers to 3 decimal places unless the question asks for an exact answer. Each part shows the highest grade it can give: [A], [M] or [E].

Quick review

- 1 Evaluate $\log_3 81$. [A]

$$3^4 = 81, \text{ so } 4$$

- 2 Solve $2^x = 50$. [A]

$$x = \frac{\log 50}{\log 2} = 5.644$$

- 3 Simplify $\frac{x^2 - 9}{x + 3}$. [A]

$$\frac{(x - 3)(x + 3)}{x + 3} = x - 3$$

- 4 Expand $(x + 5)^2$. [A]

$$x^2 + 10x + 25$$

QUESTION ONE

- (a) Solve $x^2 + 2x - 24 = 0$. [A]

$$(x + 6)(x - 4) = 0, \quad x = -6 \text{ or } x = 4$$

- (b) Solve $2x^2 - 3x - 4 = 0$. [A]

$$x = \frac{3 \pm \sqrt{(-3)^2 - 4(2)(-4)}}{2(2)} = \frac{3 \pm \sqrt{41}}{4}$$

$$x = 2.351 \text{ or } x = -0.851$$

- (c) Solve $x(x - 5) = 14$. [A]

$$x^2 - 5x - 14 = 0, \quad (x - 7)(x + 2) = 0, \quad x = 7 \text{ or } x = -2$$

- (d) Sam answered two questions. **Sam's answer has 2 errors.** Find each one, fix it, and say why it loses marks. [M]

Sam's answer

(i) Solve $2x^2 + 5x = 3$. $x(2x + 5) = 3$, so $x = 3$ or $2x + 5 = 3$, $x = -1$

(ii) How many real roots has $x^2 - 6x + 9 = 0$? $36 - 36 = 0$, so no real roots

1. (i) set to zero first: $2x^2 + 5x - 3 = (2x - 1)(x + 3) = 0$, $x = 0.5$ or -3 (L19)

2. (ii) $\Delta = 0$ means one repeated root, $x = 3$, not none (L20)

(e) Solve $(x^2 - 4x)^2 - 17(x^2 - 4x) + 60 = 0$. [E]

Let $u = x^2 - 4x$: $u^2 - 17u + 60 = 0$, $(u - 5)(u - 12) = 0$

$x^2 - 4x - 5 = 0$: $(x - 5)(x + 1) = 0$, $x = 5$ or $x = -1$

$x^2 - 4x - 12 = 0$: $(x - 6)(x + 2) = 0$, $x = 6$ or $x = -2$

u: (a), (b), (c) correct. r: (d) both errors fixed with reasons; (e) the quadratic in u solved.

t: (e) all four values of x . N1 partial; N2 1u; A3 2u; A4 3u; M5 1r; M6 2r; E7 t, minor error; E8 t.

QUESTION TWO

(a) Write an equation $x^2 + bx + c = 0$ with roots -3 and 5 . [A]

$(x + 3)(x - 5) = 0$, $x^2 - 2x - 15 = 0$

(b) Write an equation with roots $\frac{2}{5}$ and -3 , with whole-number coefficients. [A]

$(5x - 2)(x + 3) = 0$, $5x^2 + 13x - 6 = 0$

(c) A parabola crosses the x -axis at -4 and 1 and passes through $(0, 8)$. Find its equation. [M]

$y = a(x + 4)(x - 1)$; $8 = a(4)(-1)$, $a = -2$

$y = -2(x + 4)(x - 1) = -2x^2 - 6x + 8$

(d) Show that the roots of $3x^2 - 10kx + 3k^2 = 0$ ($k \neq 0$) are $3k$ and $\frac{k}{3}$. [M]

$(3x - k)(x - 3k) = 0$, so $x = \frac{k}{3}$ or $x = 3k$

(e) $x^2 + bx + 18 = 0$ has one root twice the other. Find the possible values of b . [E]

Let the roots be r and $2r$: $(x - r)(x - 2r) = x^2 - 3rx + 2r^2$

So $2r^2 = 18$, $r = 3$ or $r = -3$, and $b = -3r$: $b = -9$ or $b = 9$

Check $b = -9$: $x^2 - 9x + 18 = (x - 3)(x - 6)$, roots 3 and 6 .

u: (a), (b) correct; (c) set up. r: (c) correct; (d) shown with the factorisation.

t: (e) both values of b with reasoning. N1 partial; N2 1u; A3 2u; A4 3u; M5 1r; M6 2r; E7 t, minor error; E8 t.

QUESTION THREE

(a) Find the discriminant of $3x^2 - 5x + 4 = 0$ and state the number of real roots. [A]

$(-5)^2 - 4(3)(4) = 25 - 48 = -23 < 0$: no real roots

(b) Find k so that $x^2 - 10x + k = 0$ has equal roots. [A]

$$(-10)^2 - 4(1)(k) = 0, \quad 100 = 4k, \quad k = 25$$

(c) Find the values of m for which $2x^2 + mx + 18 = 0$ has no real roots. [M]

$$m^2 - 4(2)(18) < 0, \quad m^2 < 144, \quad -12 < m < 12$$

(d) $kx^2 + 6x + (k + 8) = 0$ has equal roots. Find k . [M]

$$36 - 4k(k + 8) = 0, \quad k^2 + 8k - 9 = 0, \quad (k + 9)(k - 1) = 0, \quad k = 1 \text{ or } k = -9$$

(e) Show that $x^2 + (2k - 1)x + k^2 - k = 0$ has two different real roots for every value of k . [E]

$$\Delta = (2k - 1)^2 - 4(1)(k^2 - k) = 4k^2 - 4k + 1 - 4k^2 + 4k = 1$$

$\Delta = 1 > 0$ whatever k is, so there are always two different real roots.

u: (a), (b) correct. r: (c) range found; (d) both values.

t: (e) $\Delta = 1$ with the conclusion for every k . N1 partial; N2 1u; A3 2u; A4 3u; M5 1r; M6 2r; E7 t, minor error; E8 t.

Totals out of 24. Cut scores move a little each year: Achievement from about 7–8, Merit 13–14, Excellence 19–20.